NumPy — LEVEL 1: Fundamentals

Beginner-friendly notes • Line-by-line

1. Introduction to NumPy

What is NumPy?

NumPy means Numerical Python. It is a Python library used for numerical calculations and working with arrays.

Why NumPy?

NumPy makes mathematical and numerical operations faster and easier than using normal Python lists.

NumPy vs Python Lists

Python lists can store different types of values. NumPy arrays are designed mainly for numerical data and usually perform calculations much faster.

Advantages of NumPy

Fast calculations, less memory usage, powerful array operations, mathematical functions, and support for multidimensional data.

NumPy applications

NumPy is used in data analysis, scientific computing, statistics, image processing, simulations, and machine learning.

NumPy in Data Science

NumPy is used to store and process numerical data. Libraries such as Pandas and SciPy are built to work closely with NumPy.

NumPy in Machine Learning

NumPy is used for numerical calculations, vectors, matrices, features, and many mathematical operations used by machine-learning algorithms.

NumPy installation

Install NumPy using pip from the terminal or command prompt.

Installation command

```
pip install numpy
```

Importing NumPy

Import NumPy so that you can use its functions in Python.

Import command

```
import numpy as np
```

Checking NumPy version

You can check the installed NumPy version using the __version__ attribute.

Version command

```
np.__version__
```

Beginner Tip: First understand what an array is, then practice creating arrays and performing basic operations on them.